



Vowels

Dr. Josef Fruehwald

2026-09-10

```
source(here::here("_extensions/mcanouil/typst-render/  
_resources/typst_define.R"))  
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —  
✓ dplyr      1.2.1      ✓ readr      2.2.0  
✓ forcats    1.0.1      ✓ stringr    1.6.0  
✓ lubridate  1.9.5      ✓ tibble     3.3.1  
✓ purrr      1.2.2      ✓ tidyr      1.3.2
```

```
— Conflicts — tidyverse_conflicts()  
—  
✖ dplyr::filter() masks stats::filter()  
✖ dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all  
conflicts to become errors
```

```
library(ggplot2)  
library(ellipsis)  
library(tidynorm)  
library(ggdensity)  
library(scales)
```



```
Attaching package: 'scales'
```

```
The following object is masked from 'package:purrr':
```

```
discard
```

```
The following object is masked from 'package:readr':
```

```
col_factor
```

```
library(geomtextpath)
```

```
library(ggdists)
```

```
reul_tri <- function(theta, n=3){  
  tau <- 2 * pi  
  f = floor(  
    ((n * (theta-pi))/tau) + 0.5  
  )  
  a = cos(theta - (tau/n)*f)  
  
  a + sqrt(1 + (2*cos(pi/n)) + (a^2))  
}
```

Vowels

What is a vowel?

- Not a consonant.

Vowels

Clues in the name

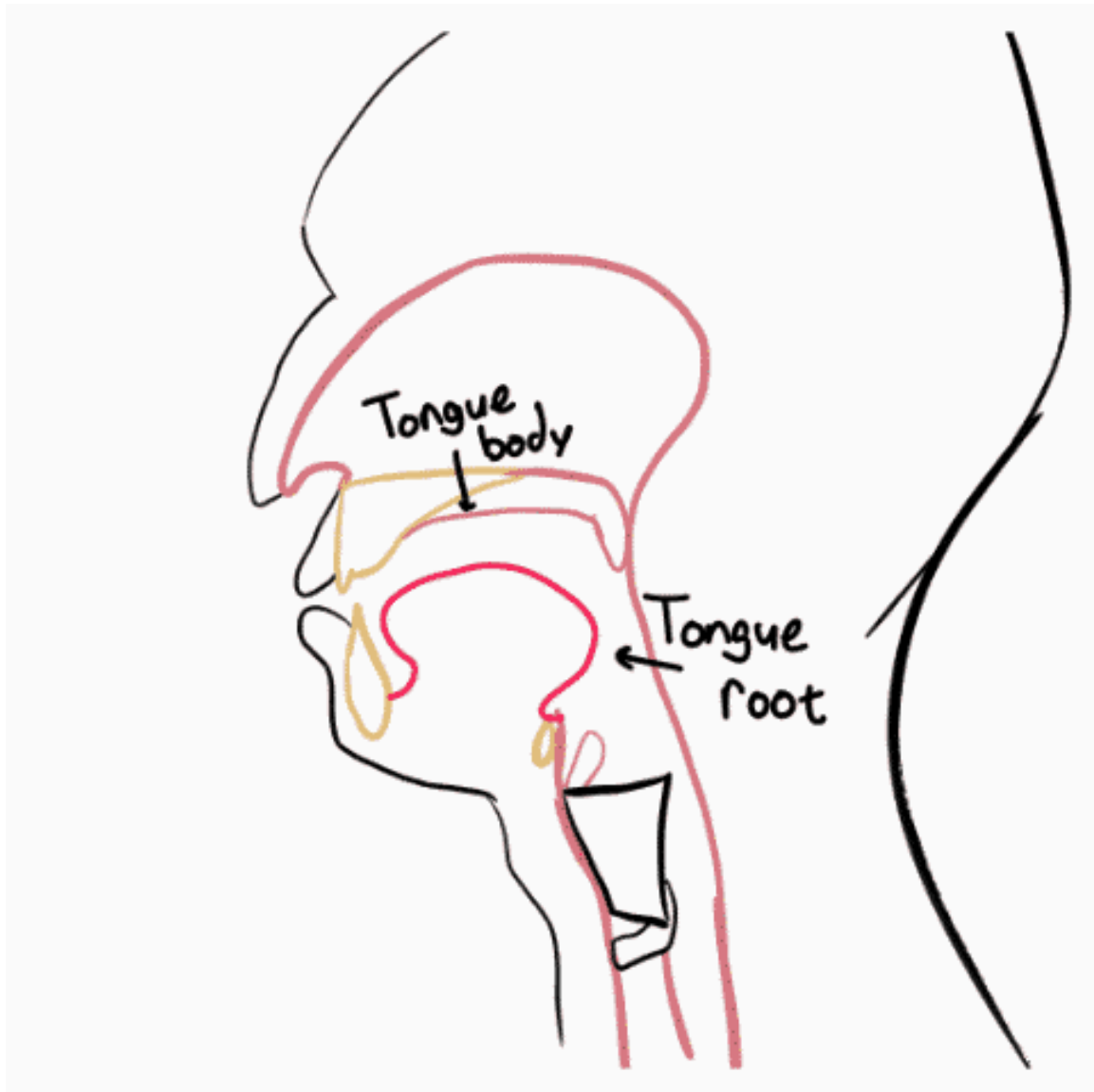
	English	Dutch
V	(sonant?)	klinker
C	consonant	medeklinker

Vowels

In general, a more open, unconstricted, articulation compared to consonants.

Vowels - Articulatory Dimensions

Tongue body and Tongue Root

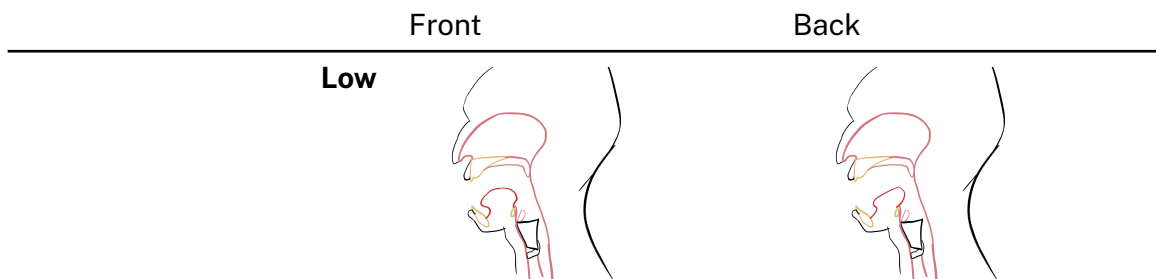


Front

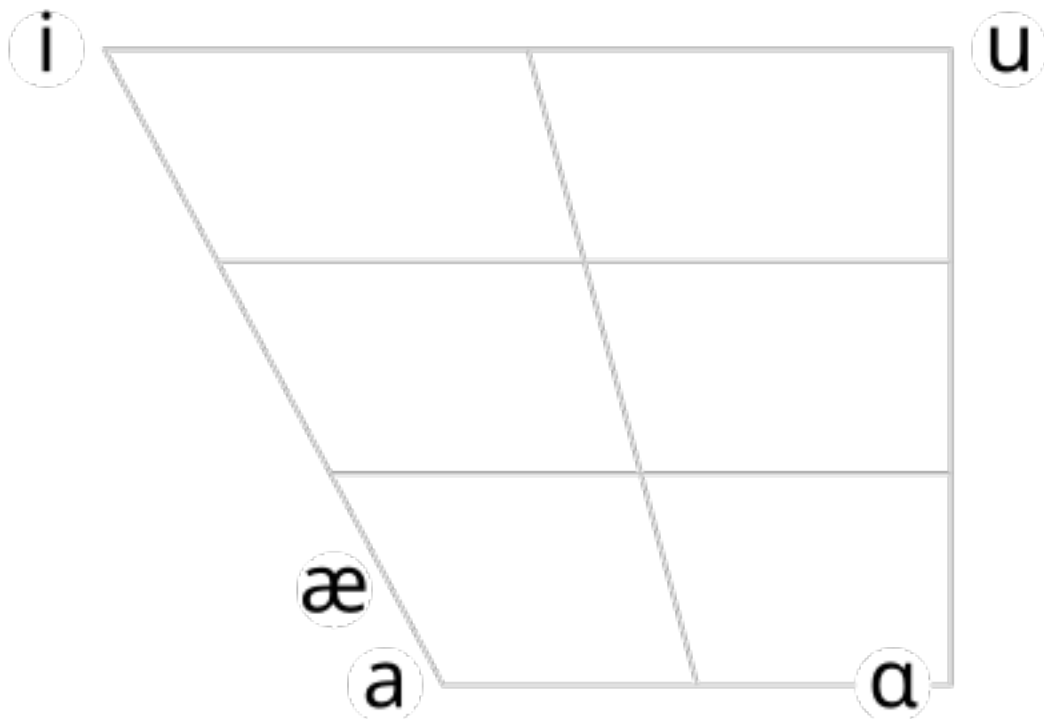
Back

High





Vowels - Dimensions



Vowels - Transcription

Vowels *fundamentally* exist in a continuous phonetic space, making the transcription problem especially tricky.

- It can be tricky putting a categorical label on a continuous space.
- Many “conventional” transcriptions use symbols that aren’t technically accurate.

Vowels - Transcription

“goose”



gU:S

?

Vowels - Transcription

```
anae <- read_rds("~/OneDrive - University of Kentucky/Corpora/ANAE/rds_data/  
anaeUNNORM.rds")
```

```
anae |>  
  norm_nearey(  
    .by = TS,  
    F1:F2  
  ) ->  
  anae_norm
```

Normalization info

- normalized with ``tidynorm::norm_nearey()``
- normalized ``F1`` and ``F2``
- normalized values in ``F1_lm`` and ``F2_lm``
- grouped by ``TS``
- within formant: FALSE
- Transformation prior to normalization: log
- `(.formant - mean(.formant, na.rm = T))/(1)`

```
anae_norm |>  
  ggplot(  
    aes(F2_lm, F1_lm)  
  ) +  
  stat_hdr(  
    probs = 0.95,  
    color = "black",  
    fill = "grey90",  
    show.legend = F  
  ) +  
  scale_x_reverse() +  
  scale_y_reverse() +  
  theme_no_x() +  
  theme_no_y() +  
  coord_fixed() ->  
  anae_base  
  
make_means <- function(df){
```



```
df |>
  summarise(
    .by = matches("^[xy]"),
    across(c(x,y), mean)
  )
}
```

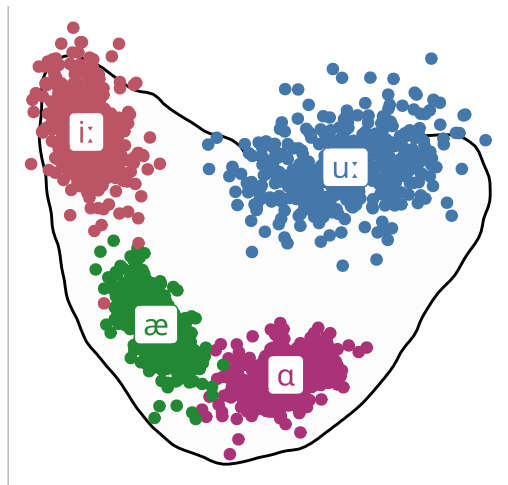
```
anae_norm |>
  summarise(
    .by = c(TS, VClass),
    across(F1_lm:F2_lm, mean)
  ) ->
  speaker_means

speaker_means |>
  filter(VClass %in% c("iy", "uw", "ae", "o")) |>
  mutate(
    vowel = recode_values(
      VClass,
      "iy" ~ "i:",
      "ae" ~ "æ",
      "uw" ~ "u:",
      "o" ~ "ɑ"
    )
  ) |>
  mutate(
    vowel = fct_reorder(vowel, F1_lm)
  ) ->
  vpoints
```

```
anae_base +
  geom_point(
    data = vpoints,
    aes(color = vowel),
    show.legend = F
  ) +
  geom_label(
    data = vpoints,
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
```



```
show.legend = F  
)
```

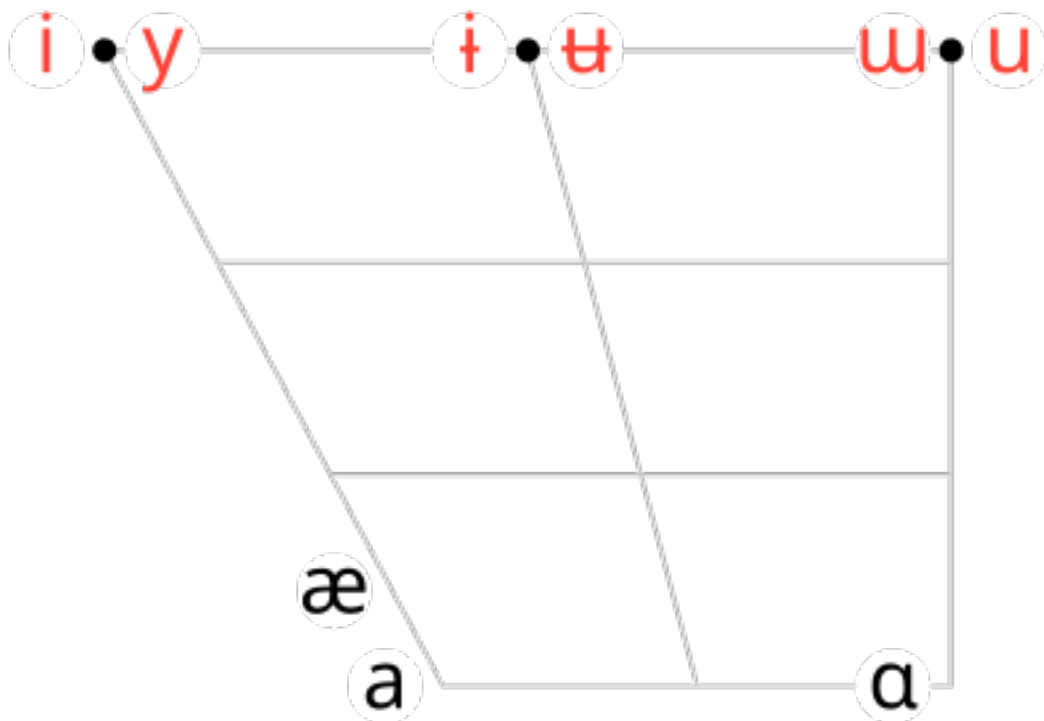


Data from the Atlas of North American English

1 point = 1 speaker

Vowels - Transcription

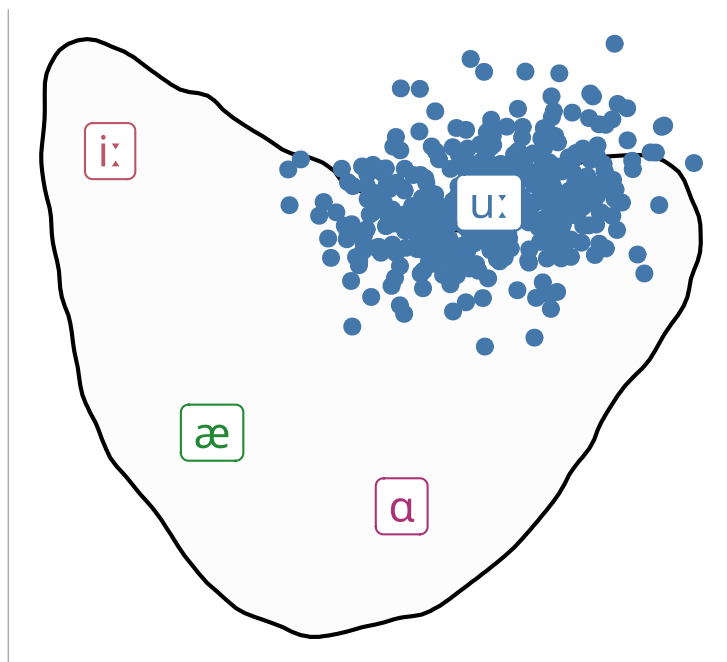
High Vowels





Vowel - Transcription

```
anae_base +  
  geom_label(  
    data = vpoints,  
    aes(color = vowel, label = vowel),  
    family = "Noto Sans",  
    stat = "manual",  
    fun = make_means,  
    show.legend = F  
  ) +  
  geom_point(  
    data = vpoints |> filter(VClass == "uw"),  
    aes(color = vowel),  
    show.legend = F  
  ) +  
  geom_label(  
    data = vpoints |> filter(VClass == "uw"),  
    aes(color = vowel, label = vowel),  
    family = "Noto Sans",  
    stat = "manual",  
    fun = make_means,  
    show.legend = F  
  )  
)
```



Vowels - Transcription

gis?

gus?

Vowels - Transcription

“thought”

θɔt

?

Vowels - Transcription

```
anae_norm |>
  filter(VClass %in% c("oh")) |>
  summarise(
    .by = c(TS, VClass),
    across(F1_lm:F2_lm, mean)
```



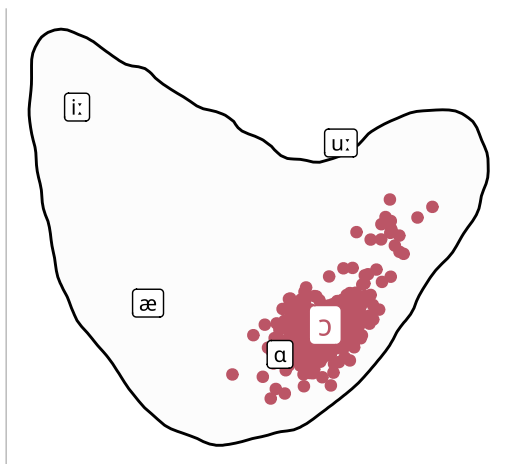
```
) |>
mutate(
  vowel = case_match(
    VClass,
    "oh" ~ "ɔ"
  )
) |>
mutate(
  vowel = fct_reorder(vowel, F1_lm)
) ->
oh
anae_base +
  geom_point(
    data = oh,
    aes(color = vowel),
    show.legend = F
  ) +
  geom_label(
    data = oh,
    aes(color = vowel, label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F
  ) +
  geom_label(
    data = vpoints,
    aes(label = vowel),
    family = "Noto Sans",
    stat = "manual",
    fun = make_means,
    show.legend = F,
    size = 4
  ) +
  theme(
    text = element_text(family = "Voces")
  )
)
```



```
Warning: There was 1 warning in
`mutate()`.
i In argument: `vowel =
case_match(VClass, "oh" ~ "ɔ")`.
Caused by warning:
! `case_match()` was deprecated in
dplyr 1.2.0.
i Please use `recode_values()`
instead.
```

Data from the Atlas of North American
English

1 point = 1 speaker



Vowels - Transcription

Back Vowels

[typst-render] Compilation failed for 'typst-block-3'.

Vowels - Transcription

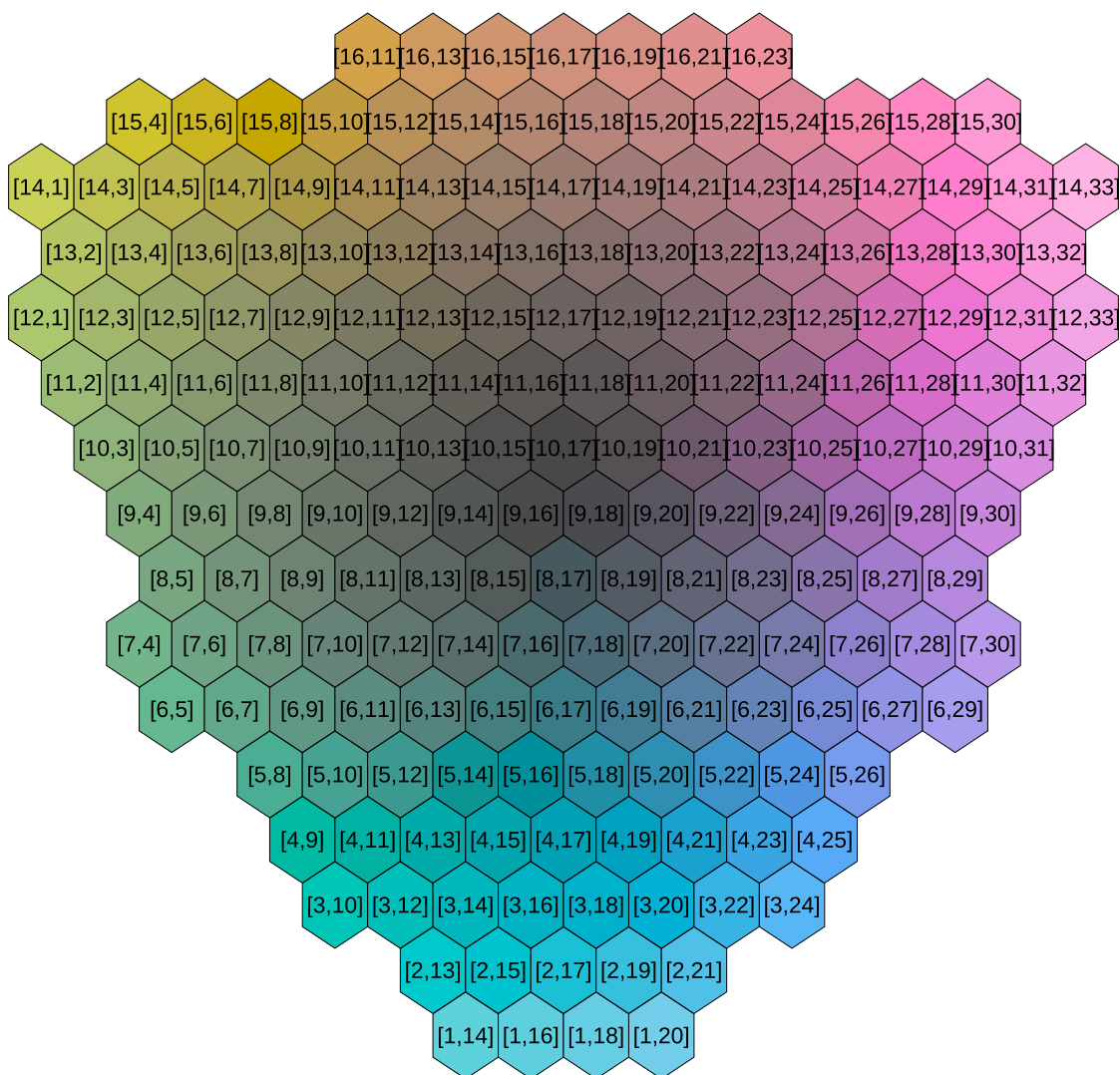
θot?

θɔt?

θɒt?



Vowels?



Vowels - Transcription

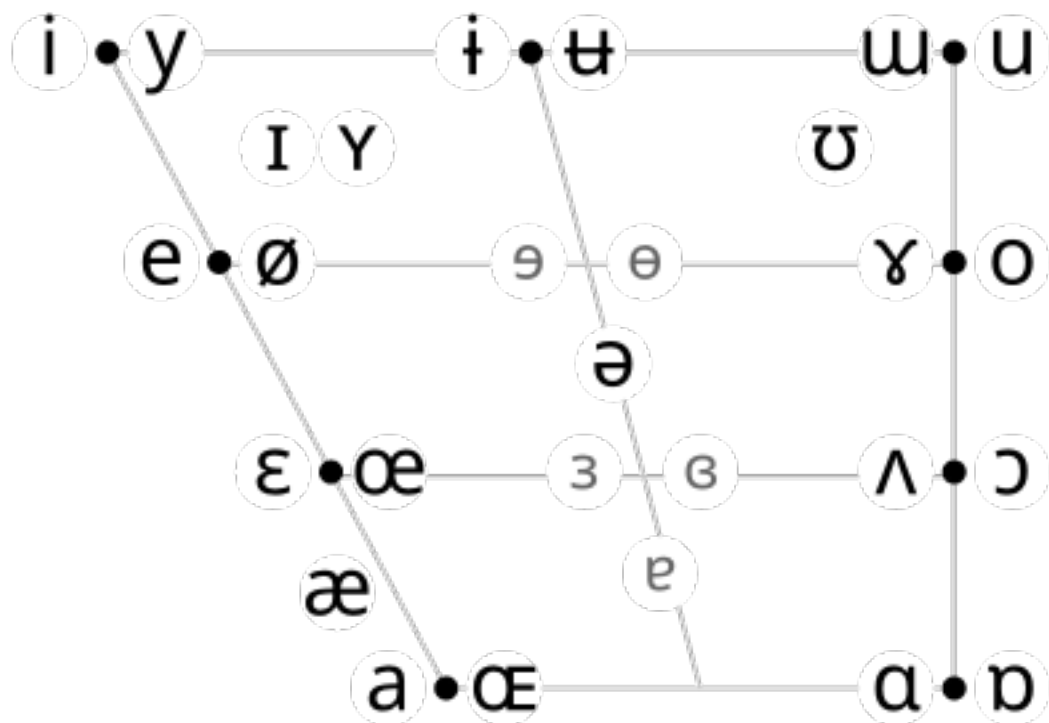
If *all* we have is paper and ink to describe phonetic detail, we should try to use the closest IPA character.

Otherwise, we'll use symbols that best characterize lexical sets.

[u:] ∈ {goose, coop, food}



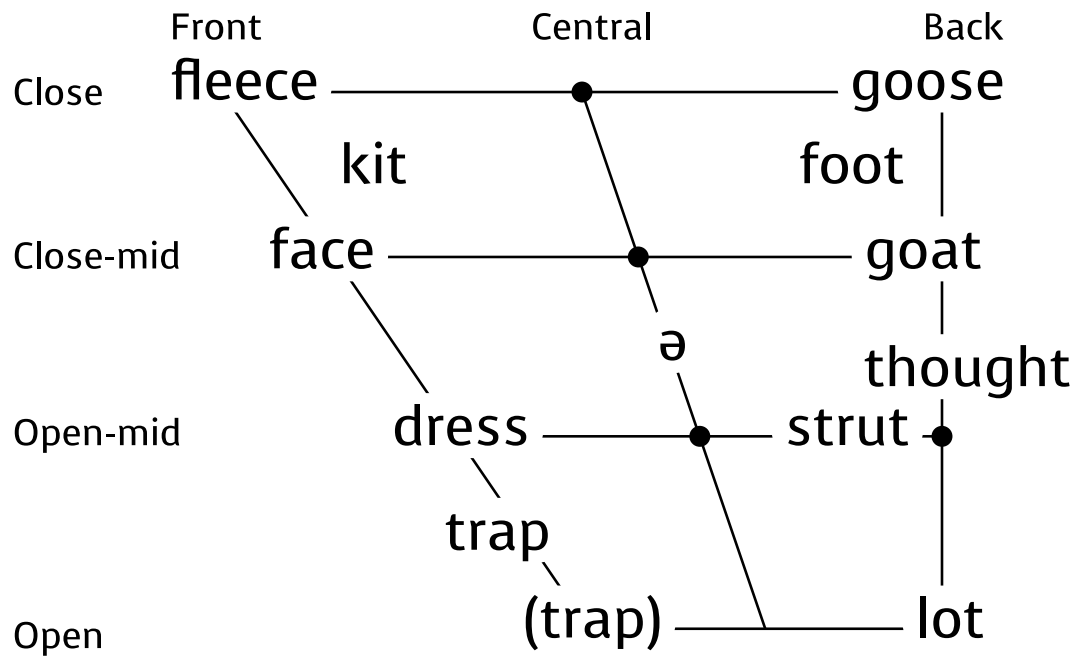
Vowels





Vowels - English Keywords

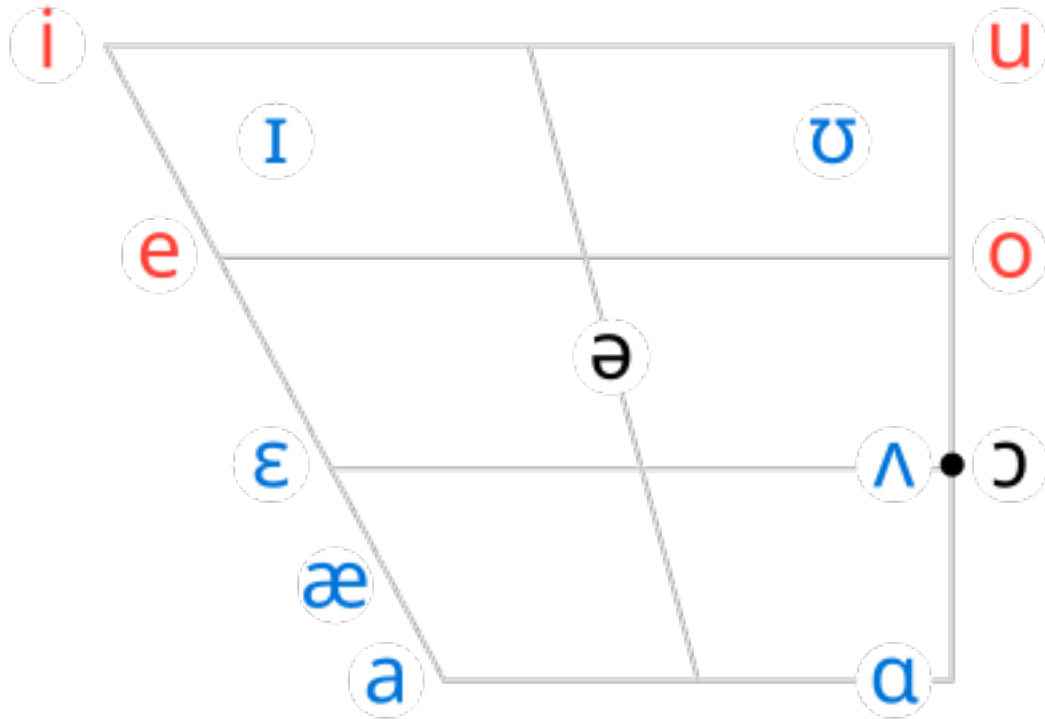
VOWELS



Vowels - Length In English



long vs short



Vowels - Length

Length is transcribed with a : symbol (which is different from :)!)

i:

u:

Vowels - Length

In English, the mid long vowels tend to be diphthongized.

Vowel - Length

The mid long vowels in English tend to be diphthongized

```
jean_lp <- read_csv("~/OneDrive - University of Kentucky/Corpora/PNC/
derived_data/fave_results/PH06-2-1-AB-Jean_logparam.csv")
```



Rows: 34635 Columns: 22

```
— Column
specification —
Delimiter: ", "
chr (12): file_name, id, group, label, word, pre_word, fol_word, pre_seg, fo...
dbl (10): param, F1, F2, F3, error, max_formant, speaker_num, optimized, str...

i Use `spec()`` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
jean_lp |>
  filter(param == 0) |>
  mutate(across(F1:F2, \(x)x*sqrt(2))) |>
  ggplot(
    aes(F2, F1)
  ) +
  stat_hdr(
    color = "black",
    fill = "grey80",
    probs = 0.95,
    show.legend = F
  ) +
  scale_x_reverse() +
  scale_y_reverse() +
  coord_fixed() +
  theme_no_x() +
  theme_no_y() ->
  jean_base
```

```
jean_lp |>
  filter(
    label %in% c("ey", "ow"),
    fol_seg != "1"
  ) |>
  summarise(
    .by = c(param, label),
    across(F1:F2, mean)
  ) |>
  reframe_with_idct(
    .param_col = param,
    .token_id_col = label,
    F1:F2
  ) |>
```

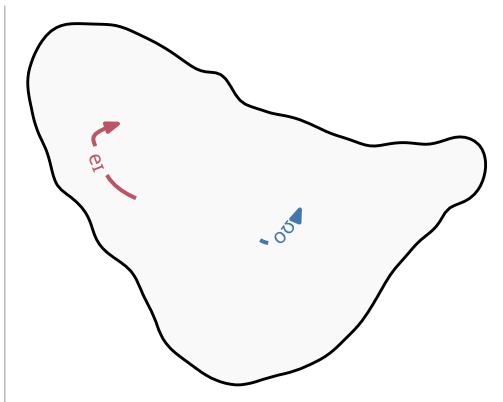


```
mutate(  
  label = case_match(  
    label,  
    "ey" ~ "er",  
    "ow" ~ "Oo"  
  )  
) ->  
jean_ey_ow  
jean_base +  
geom_textpath(  
  data = jean_ey_ow,  
  aes(  
    color = label,  
    label = label  
  ),  
  linewidth = 1,  
  arrow = arrow(  
    type = "closed",  
    length = unit(0.25, "cm"),  
    angle = 20  
  ),  
  family = "Noto Sans",  
  show.legend = F  
)
```



```
Warning in reframe_with_idct(summarise(filter(
  label %in% c("ey", : There is was no
  grouping provided.
  i You may want to provide `.by` with
  a speaker id column.
```

Average [eɪ] and [oʊ] from one speaker
from Philadelphia.



Vowels - Diphthongs

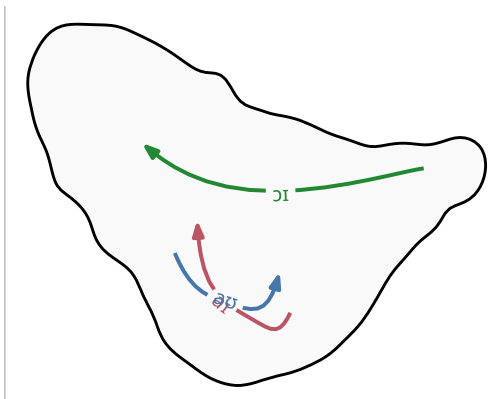
```
jean_lp |>
  filter(
    label %in% c("ay", "aw", "oy")
  ) |>
  summarise(
    .by = c(param, label),
    across(F1:F2, mean)
  ) |>
  reframe_with_idct(
    .param_col = param,
    .token_id_col = label,
    F1:F2
  ) |>
  mutate(
    label = case_match(
      label,
      "ay" ~ "aɪ",
      "aw" ~ "aʊ",
      "oy" ~ "ɔɪ"
    )
  )
```



```
)->  
jean_diph
```

Warning in reframe_with_idct(summarise(filter(jean_lp, label %in% c("ay", : There is was no grouping provided.
i You may want to provide `.by` with a speaker id column.

```
jean_base+  
  geom_textpath(  
    data = jean_diph,  
    aes(  
      color = label,  
      label = label  
    ),  
    linewidth = 1,  
    arrow = arrow(  
      type = "closed",  
      length = unit(0.25, "cm"),  
      angle = 20  
    ),  
    family = "Noto Sans",  
    show.legend = F  
  )
```

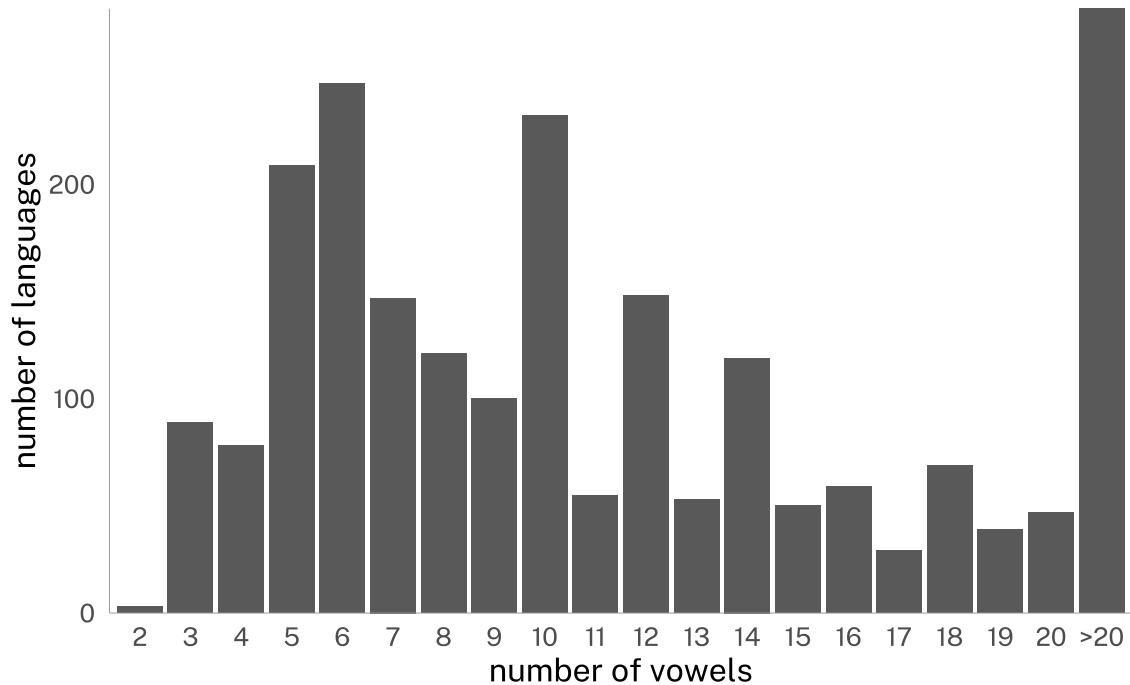


Three diphthongs from a speaker from Philadelphia



Across the world

Vowel Inventories



Inventories

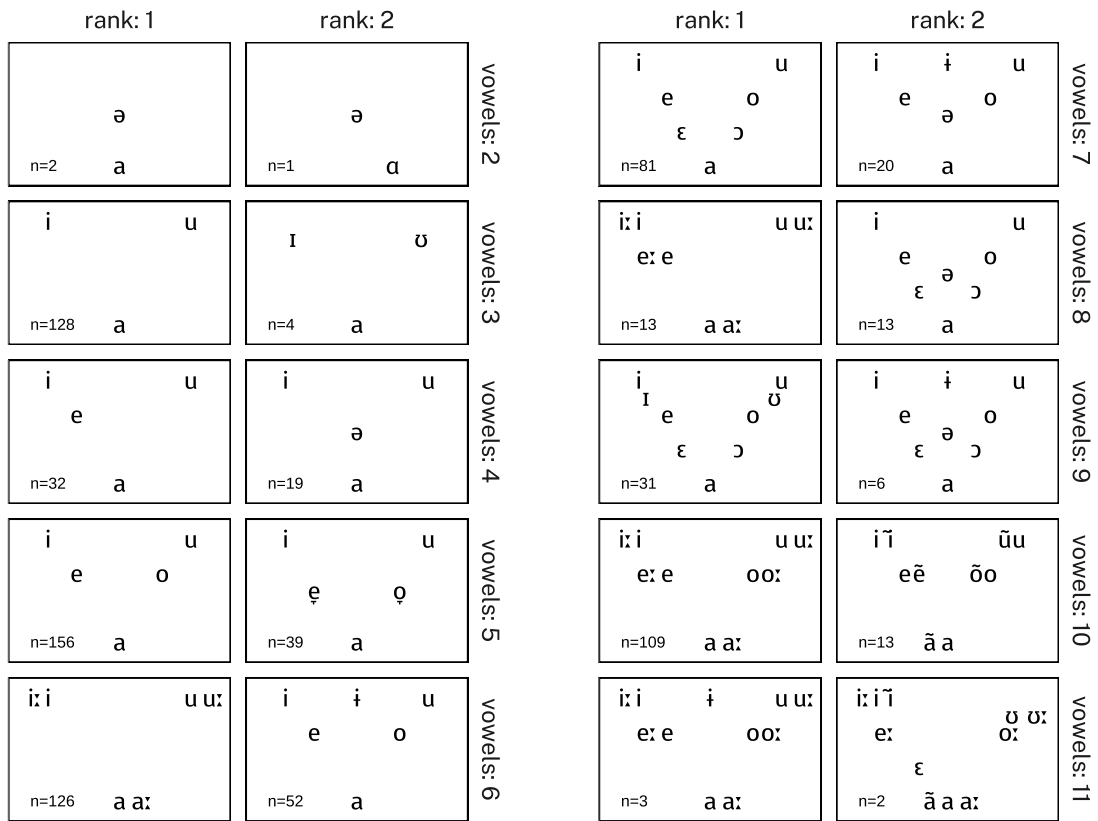
```
phoible |>
  filter(
    SegmentClass == "vowel"
  ) |>
  mutate(
    .by = c(Glottocode, SpecificDialect),
    vowels = n()
  ) |>
  count(
    vowels, Phoneme
  ) |>
  arrange(
    desc(n)
  ) |>
  slice(
    .by = vowels,
    1:first(vowels)
  ) |>
```



```
arrange(vowels) ->  
vowel_counts
```

```
tribble(  
  ~Phoneme, ~y, ~x,  
  "a", 0, 1,  
  "a", 0, 0,  
  "a:", 0, 0.6,  
  "ã", 0, -0.5,  
  "æ", 0, -0.7,  
  "e", 2, -1.2,  
  "e:", 2, -1.8,  
  "ẽ", 2, -0.8,  
  "ø", 2, -0.8,  
  "i", 3, -2,  
  "i:", 3, -2.4,  
  "y", 3, -1.6,  
  "ĩ", 3, -1.6,  
  "o", 2, 1.2,  
  "o:", 2, 1.7,  
  "õ", 2, 0.8,  
  "ý", 2, 0.8,  
  "u", 3, 2,  
  "u:", 3, 2.6,  
  "ũ", 3, 1.6,  
  "ư", 3, 1.6,  
  "ɔ", 1, 0.8,  
  "ə", 1.5, 0,  
  "ɛ", 1, -0.8,  
  "œ", 1, -0.4,  
  "ɪ", 2.5, -1.8,  
  "ʏ", 2.5, -1.4,  
  "ʊ", 2.5, 1.8,  
  "ʊ:", 2.5, 2.5,  
  "ɐ", 1.5, -1.2,  
  "ɔ̃", 1.5, 1.2,  
  "i", 3, 0,  
  
  # "ư", "ɪ", "ʊ",  
) ->  
vowel_coords
```

```
Joining with `by = join_by(Phoneme)`
```

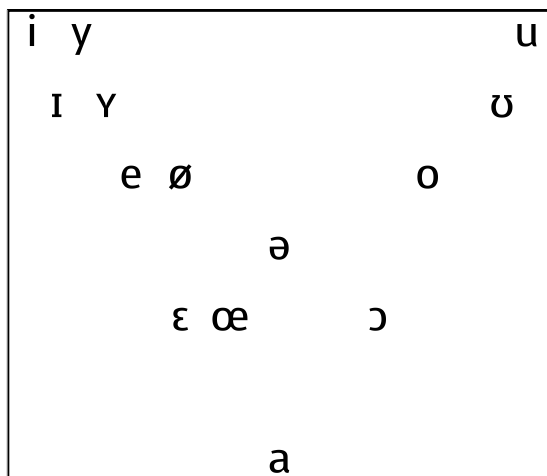


Inventories

Joining with ``by = join_by(Phoneme)``



Muyang



ngwe

